

NAOSHER MUSTAKIM

Chattogram, Bangladesh

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Dedicated and curious researcher with a passion for advancing knowledge and making meaningful contributions. Eager to drive impactful discoveries through rigorous experimentation, data analysis, and critical thinking.

Academic Credentials

University of Dhaka (DU) May 2023 – Oct 2024
M.S. in Biomedical Physics & Technology — GPA: 3.86/4.00 Dhaka, Bangladesh

Chittagong University of Engineering & Technology (CUET) Mar 2017 – Aug 2022
B.Sc. in Electrical and Electronics Engineering — CGPA: 2.99/4.00 Chattogram, Bangladesh

- CGPA: — 3.22 in last 60 credits

Publications

1. Muntaha, S., Salam, S.S., and Mustakim, N. (2024). “An Explainable AI-based Deep Learning Model for Classification of Diabetic Retinopathy Stages Using Retinal Fundus Images.” *27th International Conference on Computer and Information Technology (ICCIT 2024)*. Published.

Standardized Test Scores

International English Language Testing System (IELTS) May 2025

Overall	Reading	Writing	Speaking	Listening
7.5	7.5	6.5	7.0	8.0

Graduate Record Examination (GRE) Sep 2026

Total	Verbal	Quantitative	Analytical Writing
316	153	163	3.5

Research Experience

Collaborative Research — Mahdy’s Research Academy May 2024 – Present
Area: Statistical Machine Learning & Deep Learning for Medical Imaging
Supervisor: Dr. M.R.C. Mahdy

- Completed an intensive research-training program in AI, machine learning, and deep learning, scoring 100% across 20+ live and recorded sessions.
- Reproduced experimental results from peer-reviewed papers and ran simulation studies using TensorFlow, PyTorch, RapidMiner, and WEKA.
- Implemented hands-on experiments in medical image analysis, computer vision, explainable AI, and AI safety, and presented journal-article critiques in group sessions.

Graduate Thesis: Deep Learning for Multi-Label Classification of Bi-manual Clench Force fNIRS Data Sept 2023 – Jul 2024
Supervisor: Dr. Muhammad Abdul Kadir

- Designed a custom force sensing device using FSR sensors and Arduino Mega.
- Built a custom graphical user interface for real-time visual feedback to participants using Python and Tkinter.
- Collected fNIRS brain-activity data from participants across five hand-clench force levels for both hands.
- Constructed, trained, and evaluated a custom multi-label deep learning classifier combining CNN, spectral attention, and LSTM layers.
- Applied explainable AI (SHAP) to identify the spectral and temporal features driving the model’s force-level predictions.

Undergraduate Thesis: Design & Fabrication of a Wideband Slot-Loaded Textile Antenna

Jan 2022 – Jul 2022

Supervisor: Dr. Muhammad Asad Rahman

- Designed a wideband textile antenna using ADS and CST simulation software.
- Fabricated the antenna using 100% polyester as substrate and copper tape as the radiating patch and ground plane.
- Achieved bandwidth of 17.36% in simulation and 11.5% in real world; peak gain of 7.81 dBi with 88% radiation efficiency.
- Achieved a Specific Absorption Rate (SAR) of 0.316 W/kg at 100 mW input power, within the FCC guideline of 1.6 W/kg.
- Applications include patient monitoring, protective suits for rescue workers, and military personnel.

Professional Experience

Engineer, Electrical — Bangladesh Steel Re-Rolling Mills Ltd (BSRM)

Mar 2024 – Present

Chattogram, Bangladesh

- Maintain electrical systems for induction furnaces and plant utilities, working across control systems and power electronics.
- Perform root-cause analysis and troubleshooting to minimize downtime in a continuous steel-production environment.

Technical Skills

Programming & Libraries: Python, PyTorch, TensorFlow, Scikit-Learn, NumPy, Pandas, Matplotlib, Plotly, Seaborn, MATLAB, C, $\LaTeX$

Software & Tools: Microsoft Office, Overleaf, Draw.io, RapidMiner, WEKA, COMSOL, ADS, CST, Origin

Volunteering Experience

Omdena

Jul 2021 – Aug 2021

Junior Machine Learning Engineer

Chattogram, Bangladesh

- Web-scraped road accident data from news sources.
- Cleaned, preprocessed, visualized, and analyzed the data.
- Built a time-series forecasting model using LSTM.

Awards

Government Technical Scholarship awarded based on merit in the undergraduate entrance examination.

Language Proficiency

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| • English — Proficient (IELTS 7.5) | • Hindi — Intermediate (Conversational) |
| • Bangla — Native | • German — Beginner |

References

Dr. Muhammad Abdul Kadir

Professor and Chairman, Department of Biomedical Physics & Technology

University of Dhaka, Dhaka – 1000, Bangladesh

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